# ResearchStudio-Reel: Automate the Last Mile of Research from Paper to Poster, Video, and Blog

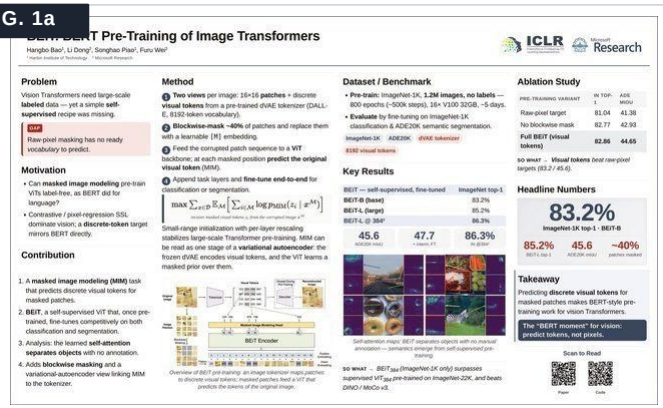
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One paper PDF in, **three editor-ready artifacts** out, and one navigable surface that binds them.

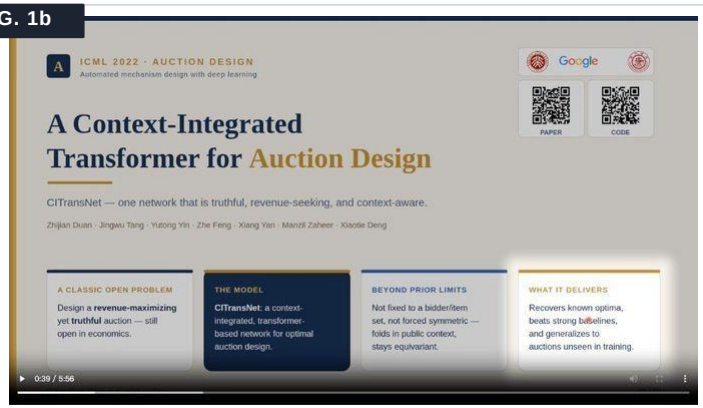
(a) Poster — Paper2Poster

FIG. 1a




(b) Talk video — Paper2Video

FIG. 1b




(c) Bilingual blog — Paper2Blog

FIG. 1c

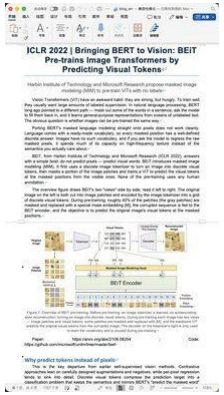



Figure 1 of the paper: three editor-ready artifacts from one paper. Project:

# The last mile of research is a separate layer, prepared in the days after acceptance, and three gaps remain open

Poster, talk video, and blog piece each need figure selection, layout, narration, or audience-appropriate writing (§1)

The dissemination layer covers the poster you stand next to at a conference session, the talk video on the venue's virtual track or the lab's channel, and the blog piece shared through social media. Authors prepare all three in the days right after acceptance, at exactly the moment they have the least time to spare; labs, courses, and multilingual organizations reuse the same layer.

> Paper-to-poster, paper-to-video, and paper-to-blog systems, plus the multi-artifact suites PaperX and OmniPresent, shift the question from whether several formats can be generated to **whether the artifacts remain native-editable and navigable** .

### G1 Shared grounding

Combining artifact-specific systems repeats figure, caption, and metadata extraction, and leaves cross-artifact references for the user to reconcile.

### G2 Native editability

A multi-format suite does not by itself give native-editable source files for every artifact in the author's PowerPoint and Word workflow.

### G3 Experience-level convergence

Parallel or consistently rendered outputs do not bind poster regions, video segments, and blog passages into one interactive surface for revision and reuse.

## WHO ELSE THE LAST MILE SERVES (§1)

**The accepted-paper author**

the poster, talk video, and blog piece prepared in the days after acceptance, right after the camera-ready

**An industrial research lab**

public-facing communications for every accepted paper, polished by a communications team

**A course or multilingual organization**

a graduate course's paper-of-the-week briefing pack; cross-region outreach in more than one language

Source: arXiv:2607.04438 v2 (19 Jul 2026)

# Prior work is organized by output type; contracts differ in editability, and VLM scores are mostly used for evaluation, not as release gates

Representative systems per output type, what they emit, and the contract each leaves open (§5.1–5.5)

| Output type | Representative systems | Output contract | What remains open |
|---|---|---|---|
| **Paper-to-poster** | | | |
| **Slides and video** | | | |
| **Blog and long-form** | | | |
| **Multi-artifact suites** | | | |

> "Rather than claiming the first multi-artifact generator, ResearchStudio-Reel focuses on a native-editable, section-aligned dissemination workspace." (§5.5) The distinction it draws is native editability plus experience-level convergence, backed by release checks.

## WHAT THE PAPER IS SCORED AGAINST IN TABLE 1 (§3)

**Single-shot frontier LLMs**

Claude-4.8 Opus, GPT-5.5, Gemini-3.1 Pro — one fixed prompt, one call, a whole A0 poster as HTML (Appendix G)

**Reproduced poster pipelines**

Paper2Poster tool, P2P, PosterGen under their released configurations, scored by the same pipeline

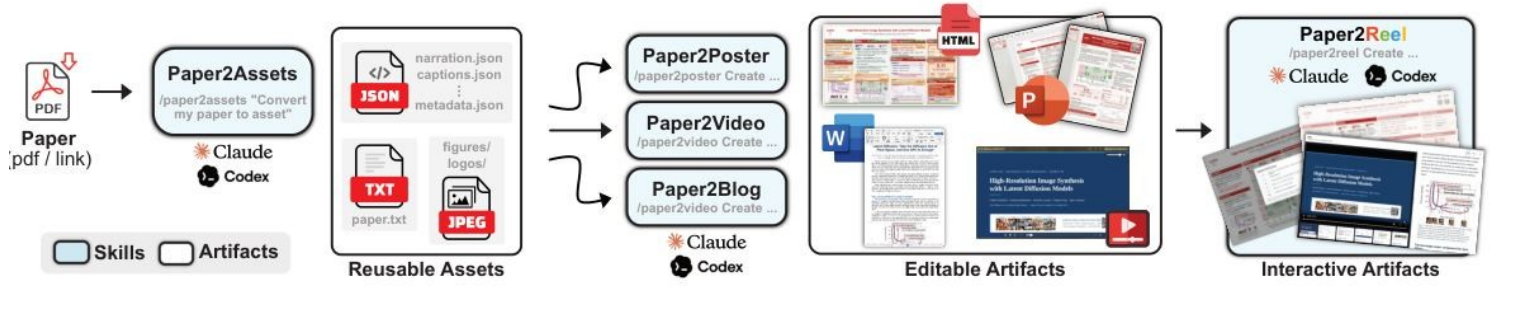
**Human reference**

The authors' own posters for the 100 benchmark papers; OmniPresent is discussed qualitatively, not scored

# The answer: a native-editable workspace of five skills — one shared extractor, three editable generators, one convergence layer

Figure 2: one PDF in, three editor-ready artifacts out, one shared extraction in the middle (§1, §2)

FIG. 2




## FIVE SKILLS, IN PIPELINE ORDER

| | | | |
|---|---|---|---|
| 1 | **Paper2Assets** | shared extractor | answers G1 |
| 2 | **Paper2Poster** | editable generator | answers G2 |
| 3 | **Paper2Video** | editable generator | answers G2 |
| 4 | **Paper2Blog** | editable generator | answers G2 |
| 5 | **Paper2Reel** | convergence layer | answers G3 |

## CONTRIBUTIONS, AS THE AUTHORS STATE THEM

1. The workspace itself: one paper PDF becomes a print-ready conference poster, a narrated talk video, and a bilingual blog through composable skills.

2. Paper2Reel, an experience-level convergence layer whose interactions connect poster regions, video segments, and blog passages in one navigable HTML surface.

3. A delivery contract: one shared Paper2Assets bundle plus native-editable sources — PowerPoint for the poster and the video deck, Word for the bilingual blog.

4. Posters best among automated systems on all three aesthetic sub-criteria and best or tied-best on two of three information sub-criteria; about 17.5% above the authors' own on aesthetics (3.56 vs 3.03); overall wins on 74 and 95 of 100 papers under the two judges. The video ships narration-aligned highlights, burned-in and sidecar subtitles, and duration control; the blog ships in two languages with layout-aware DOCX repair.

Runtimes: Claude Code and Codex skill mechanisms; deterministic primitives — headless Chromium (HTML→PDF), LibreOffice + ffmpeg (slides→video), python-docx (.docx); Edge TTS for narration. The runtime itself is not a contribution.

Source: arXiv:2607.04438 v2 (19 Jul 2026)

# Paper2Assets reads the PDF once and emits one bundle every generator consumes verbatim; figure cleanup runs once

The bundle is the only interface; stable ids, figure handles, and claim anchors keep the artifacts aligned (G1)

| Text, captions, figures | Metadata | Nine-section summary | Logos and QR | Canonical records | Second pass |
|---|---|---|---|---|---|
| pull text and captions; crop each figure with a column-aware margin | title, authors, institutes, venue, paper and code links from page 1 / arXiv page | Problem … Takeaway; each with an essential entry, a reserve entry, and a spoken script | institution marks from Wikimedia infoboxes, venue from Wikidata; slot hidden when unresolved | emit the record files the downstream skills read | inventory manifest with PDF checksum, per-section ids, narration list in reading order |

## FIGURE-CLEANUP CHAIN — THE LOAD-BEARING STAGE, RUN ONCE PER FIGURE

**Deterministic prefix**
strips chrome residue, baked-in caption strips, uniform margins

→

**Visual-AI crop**
judges a tight bounding box for the hard cases

→

**Fresh-context verifier**
a sub-agent re-reads the original against the proposed crop

→

**Commit only on a clean pass**
splits a raster that packs two figures; raw extract backed up once

## WHAT THE BUNDLE HOLDS

- Full body text with page breaks; detected figure captions; a per-figure manifest; cleaned figure images
- Metadata: title, authors, institutes, venue, paper and code links; best-effort institution and venue logos; QR codes
- The nine-section summary — Problem, Motivation, Contribution, Method, Dataset/Benchmark, Key Result, Ablation, Headline Numbers, Takeaway — plus the inventory manifest, per-section records with stable ids, and the narration clip list

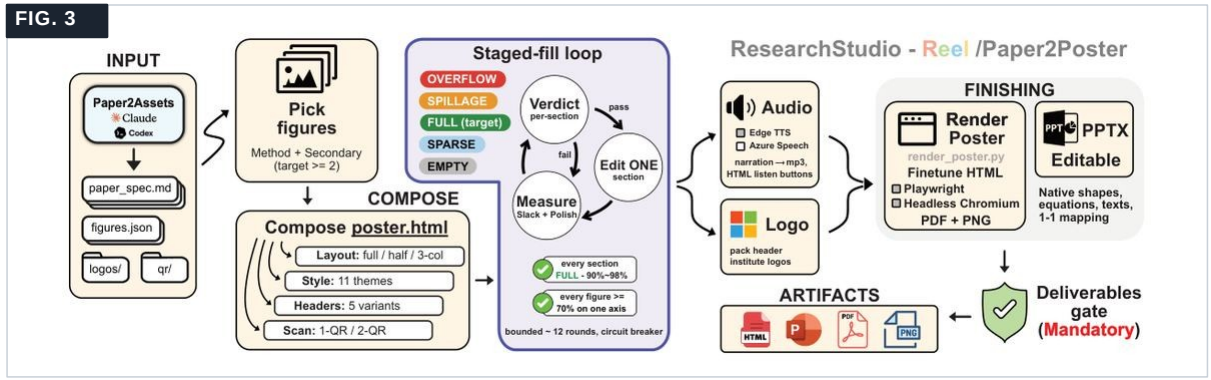
### Single-owner extraction

Downstream skills read the bundle and never re-open the PDF. A stable figure-naming invariant means the figure the poster's method card embeds is the one the video's slide cites and the blog's evidence map references, without any cross-skill coordination. Each step is an idempotent script, so a failed stage re-runs in isolation.

# Paper2Poster composes on four orthogonal axes, fills the page with a measured loop, and exports native PowerPoint shapes

Figure 3: bundle → pick figures → compose poster.html → staged-fill loop → PDF, PNG, editable PPTX → deliverables gate

FIG. 3



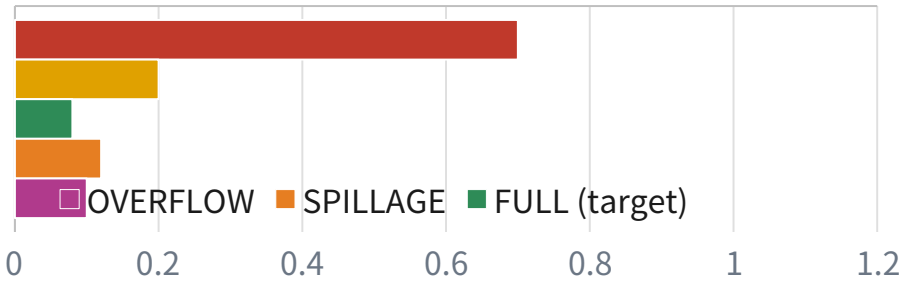

## ONE RUN, FOUR COUPLED FORMATS

- A self-contained web page (the visual ground truth)
- A print-resolution PDF at the exact venue canvas (60 × 36 in landscape for ICML / NeurIPS / CVPR, A0 portrait for the ACL family)
- A thumbnail PNG and a natively editable PowerPoint file
- Optional per-section narration audio (Edge TTS or Azure Speech)

| ID | Requirement | Solution the paper gives |
|---|---|---|
| A1 | | |
| A2 | | |
| A3 | | |
| A4 | | |
| A5 | | |

# The fill loop is discrete: five verdict bands, one section edited per round, termination as a categorical fixed point

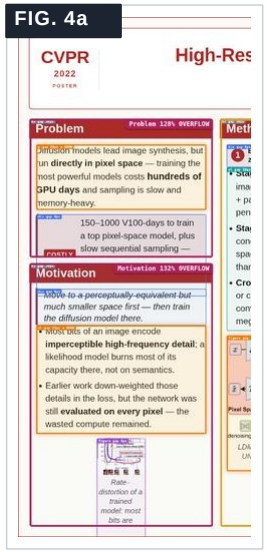
fullRatio = content height ÷ card height; the verdict alone selects the move; Figure 4 shows one poster converging



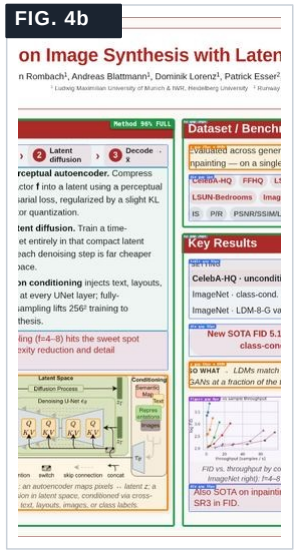

| | |
|---|---|
| **EMPTY below 0.70** | append a reserve paragraph or promote an optional section |
| **SPARSE 0.70–0.90** | polish up: pad prose or enlarge a widget to reach the card floor |
| **FULL 0.90–0.98** | the target band, left untouched |
| **SPILLAGE 0.98–1.10** | polish down: tighten prose until it re-enters the band |
| **OVERFLOW above 1.10** | drop a reserve paragraph or an entire optional section |

## WHY IT TERMINATES

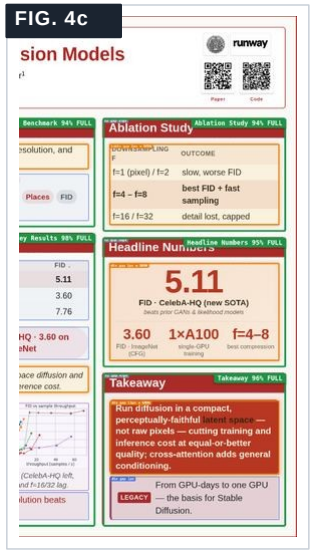
- Each move is sized by the signed pixel delta the measurement reports, never guessed
- The loop refuses to re-apply a move that already overshot the band on a section
- An on-disk round counter trips a circuit breaker (bounded ≈ 12 rounds) and ships the best-measured state
- Exit: every section reads FULL and no figure trips the figure-fill gate (≥ 90% of its card on one axis); a render-time pass then stretches under-filled cards toward 98% by growing row whitespace, figures not resized




(a) Initial explorations




(b) Fill loop in progress




(c) Poster completed

Debug overlay: every section boxed and colored by verdict (red / amber for EMPTY / SPARSE, green for FULL, orange / magenta for SPILLAGE / OVERFLOW), annotated with its fill percentage.

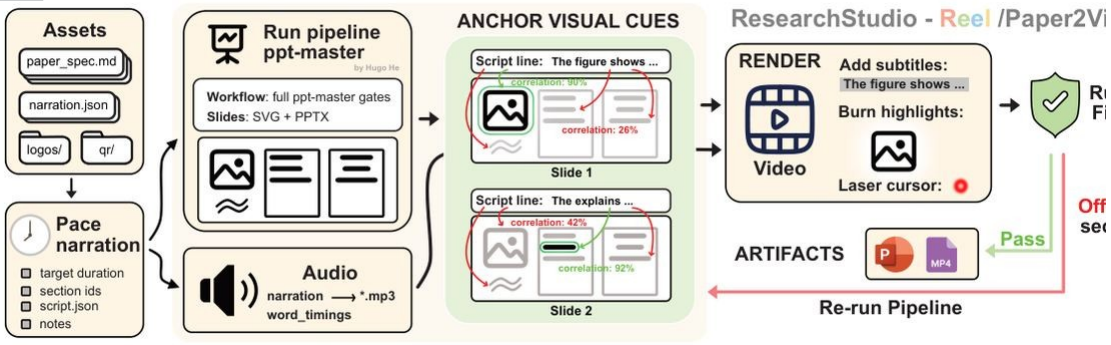
| | | | |
|---|---|---|---|
| **90%** | **98%** | **≈12** | **90%** |
| section fill gate | expand target | rounds, then breaker | figure-fill gate |

Source: arXiv:2607.04438 v2 (19 Jul 2026)

# Paper2Video plans duration before rendering, anchors narration to visible objects, and gates an editable deck plus two MP4s

Figure 5: assets → pace narration → deck via the full ppt-master workflow → anchored visual cues → render → fit gate

FIG. 5




## ROOT DELIVERABLES AND THE PACKAGE GATE

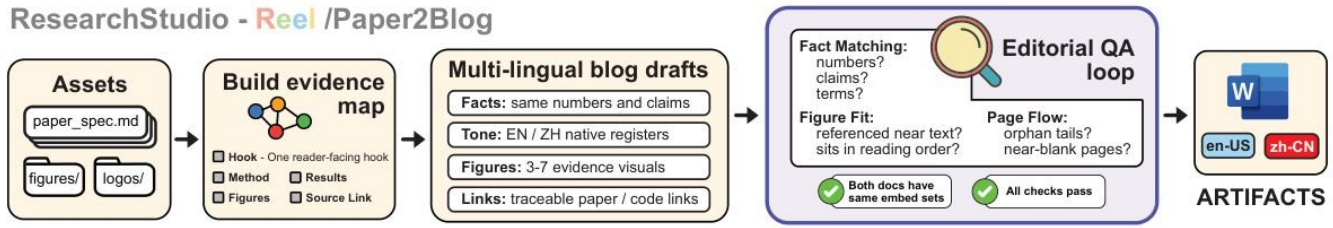
- video.pptx — the editable deck, produced by the full ppt-master route [22]; not a disposable intermediate
- video.mp4 with burned-in subtitles and video_no_subtitles.mp4 for the reel's own caption toggle; same frames, audio, highlight timing
- assets/ keeps narration audio, subtitle sidecars, frames, duration reports, cue plans, timeline.json, QA reports
- check_video_package.py is the release contract: files exist and play, audio stream present, duration in tolerance, non-empty sidecar, cues timed inside their audio segment, no word-sized highlight boxes, blank frames and text overflow caught, unsafe rate changes rejected

| ID | Requirement | Solution the paper gives |
|---|---|---|
| B1 | | |
| B2 | | |
| B3 | | |
| B4 | | |
| B5 | | |

# Paper2Blog writes the Chinese and English articles separately from one evidence map, then gates the Word files on layout

Figure 7: assets → evidence map → multilingual drafts → editorial QA loop → the Chinese and English DOCX (§2.4)

FIG. 7




## ROOT DELIVERABLES

- The Chinese article, cast as a WeChat public-account piece
- The English article, cast as a research-blog piece
- Same facts, figures, numbers, claims, and source links; different openings, pacing, and examples; outlines, previews, and QA reports stay in the intermediates directory

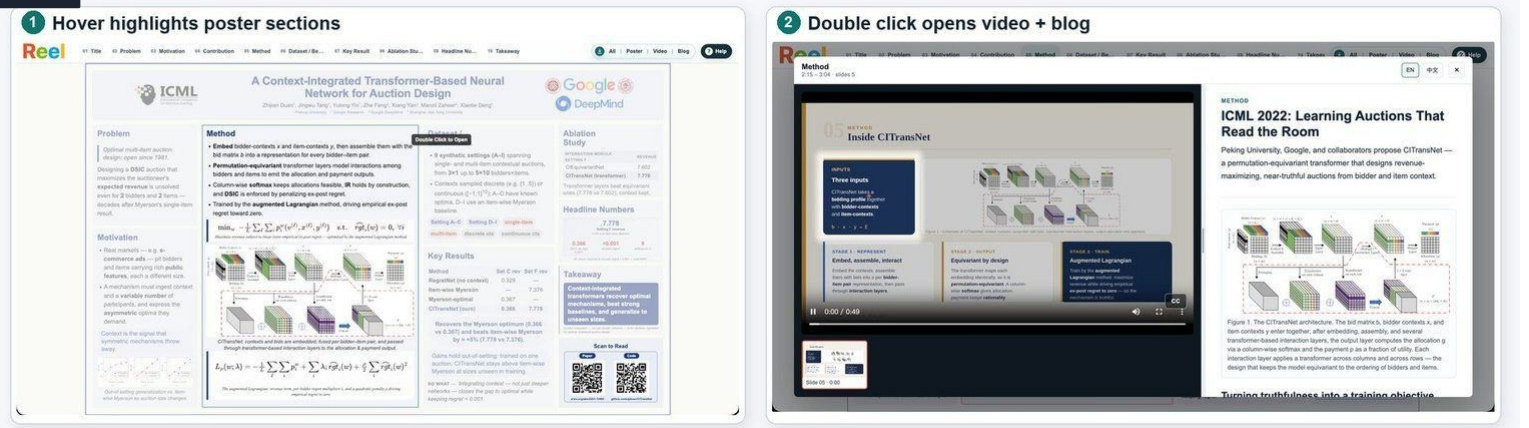
| ID | Requirement | Solution the paper gives |
|---|---|---|
| C1 | | |
| C2 | | |
| C3 | | |
| C4 | | |
| C5 | | |

Package gate: both DOCX exist and are readable, contain enough article text, embed images rather than link them, carry no placeholder text, declare a Latin body font and a Chinese fallback; embedded image count, identity, and order are compared across the two languages, extracted numeric claims and technical-looking terms are cross-checked — concrete divergences, not writing style.

# Paper2Reel is a presentation surface, not a fourth summary: one alignment sidecar makes four artifacts views of one section

Figure 9: hover highlights a section; double-click opens video, captions, thumbnails, and the aligned blog passage

FIG. 9




## INTERACTION DESIGN

- Poster-first: the first screen is the generated poster, not a dashboard
- Hover feedback; double-click opens the section modal — video left, blog right, draggable splitter
- Playback uses the subtitle-free source; captions come from sidecars behind the viewer's toggle
- Slide thumbnails seek to their time; the title area opens a full-paper modal; EN / 中文 switch in the header

**Bundle root, kept to three files**

The interactive viewer page, the alignment record, and the package manifest; poster, media, slides, blog, and downloads sit in their own folders and stay downloadable as native files.

## CONTENT ALIGNMENT

- The alignment sidecar maps each canonical section id to its poster block, one or more slide targets, video start and end, subtitle tracks, thumbnails, and blog blocks; the viewer never scrapes filenames, the poster, or the video
- On a PDF, an arXiv link, or a partial bundle, missing upstream stages are completed through the full workflows, preserving section ids and asset paths
- Root: the viewer page, the alignment record, the manifest; poster, media, slides, blog, and downloads in their own folders; native files stay downloadable

## VALIDATED AS AN INTERACTIVE ARTIFACT

- Static gate: viewer, alignment record, manifest, copied poster assets, media, slide frames, blog blocks, wordmark, downloads all present; no stale tabbed-viewer markers, machine-local paths, or backup files; raw pre-subtitle video as playback source
- Browser gate: served with a range-capable local server (seeking and thumbnail jumps need partial-content responses); headless checks of poster-first load, hover, both modals, split pane, subtitle toggle, thumbnail and direct seeking, menu, downloads, shortcuts, blog rendering
- A separate file-browser gate confirms the page also opens directly from disk

# Best automated system on all three aesthetic sub-criteria (3.56 vs the authors' 3.03) and on two of three information sub-criteria

Table 1: 100-paper Paper2Poster benchmark; cells average two VLM judges (claude-opus-4.8, gpt-5.5); PaperQuiz is exact-match %

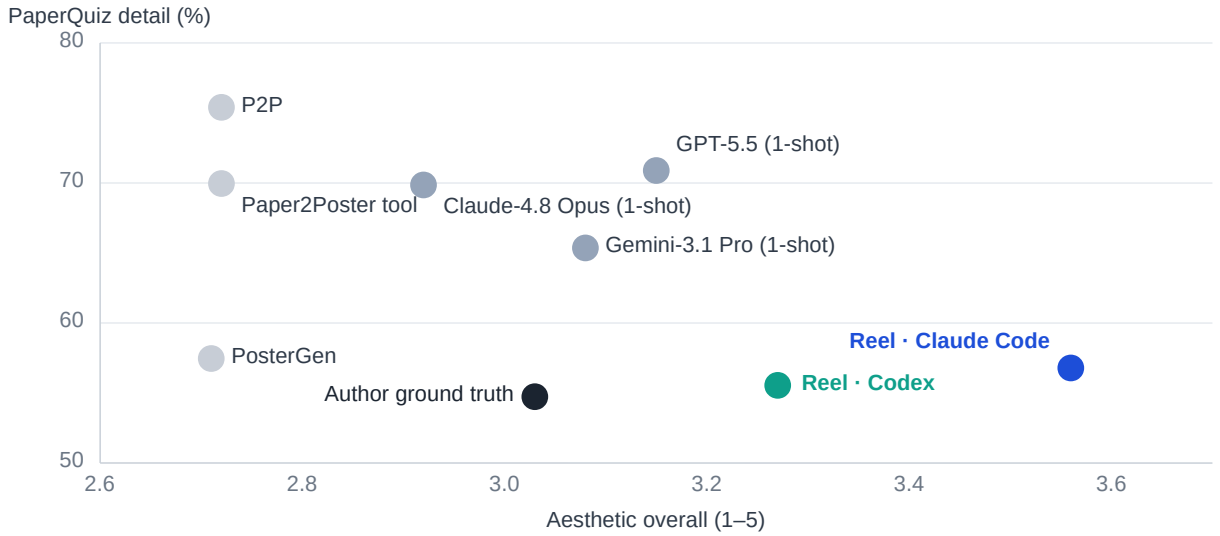
| | AESTHETIC (1–5) | | | INFORMATION (1–5) | | | | PAPERQUIZ (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **System** | **Elem.** | **Engag.** | **Layout** | | **Low** | **Logic** | **Cont.** | | | |
| Claude-4.8 Opus (single-shot) | 3.10 | 2.68 | 2.97 | 2.92 | 3.92 | 4.01 | 3.80 | 3.91 | 69.86 | 93.67 |
| GPT-5.5 (single-shot) | 3.12 | 2.94 | 3.40 | 3.15 | **3.99** | 4.00 | **3.91** | **3.97** | 70.89 | 94.25 |
| Gemini-3.1 Pro (single-shot) | 3.07 | 2.89 | 3.27 | 3.08 | 3.95 | 4.00 | 3.65 | 3.87 | 65.36 | 93.59 |
| Paper2Poster tool [1] | 2.86 | 2.30 | 3.00 | 2.72 | 3.46 | 3.41 | 3.34 | 3.40 | 69.95 | 95.02 |
| P2P [3] | 2.68 | 2.34 | 3.15 | 2.72 | 3.92 | 3.78 | 3.70 | 3.80 | **75.40** | **95.60** |
| PosterGen [2] | 2.93 | 2.23 | 2.98 | 2.71 | 3.80 | 3.76 | 3.37 | 3.64 | 57.45 | 91.85 |
| ResearchStudio-Reel (Codex, gpt-5.5) | 3.16 | 3.03 | 3.62 | 3.27 | 3.95 | 3.97 | 3.09 | 3.67 | 55.55 | 92.01 |
| ResearchStudio-Reel (Claude Code, claude-opus-4.8) | **3.53** | **3.17** | **3.99** | **3.56** | **3.99** | **4.03** | 3.41 | 3.81 | 56.79 | 92.16 |
| Author ground truth (human reference) | 3.13 | 2.60 | 3.35 | 3.03 | 3.45 | 3.85 | 3.50 | 3.60 | 54.73 | 91.40 |

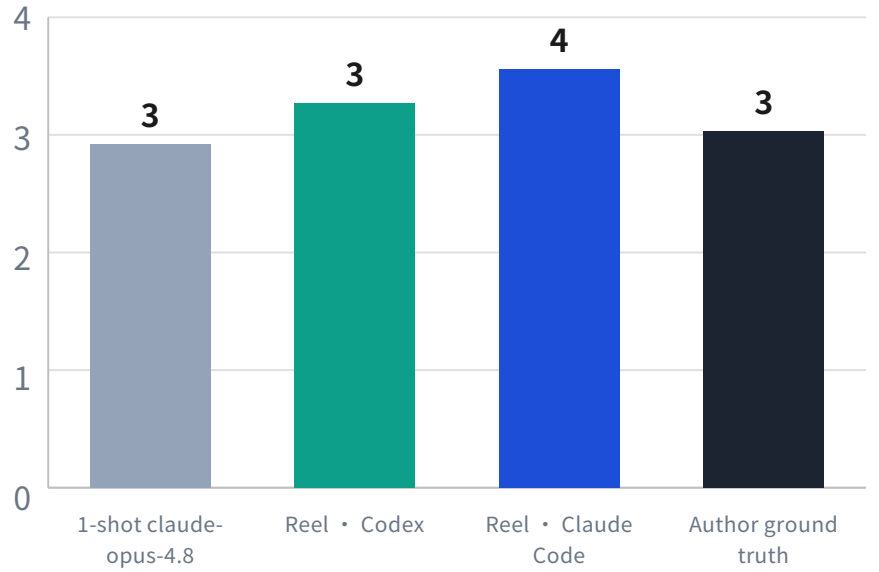
- Overall score (mean of the two Overall columns): Claude Code leads at 3.69; single-shot LLMs 3.41–3.56 and Codex 3.47 form a second tier; reproduced pipelines 3.06–3.26. Bold = best per column among the systems.
- Per paper, Claude Code beats the authors' poster on overall score for 74 of 100 papers under the Claude judge and 95 under the GPT judge (85% and 100% of non-tied papers). On PaperQuiz the ranking inverts (next page).

Source: arXiv:2607.04438 v2 (19 Jul 2026)

# The PaperQuiz ordering nearly reverses the aesthetic ordering: denser posters answer more; the fill loop packs to a target density

Left: aesthetic overall vs PaperQuiz detail, nine systems; right: same model with and without the loop, same skill under Codex




- P2P leads both Quiz splits: its portrait canvas is packed with near-verbatim prose, at some of the lowest aesthetic scores. The Paper2Poster tool is close behind because its content is assembled from the benchmark's own Q&A extraction. Author posters prune to headline results and place last on both Quiz splits.
- The single-shot baselines are strong on content and intermediate on aesthetics; Reel sits deliberately between the poles.

**The authors' caution, kept**

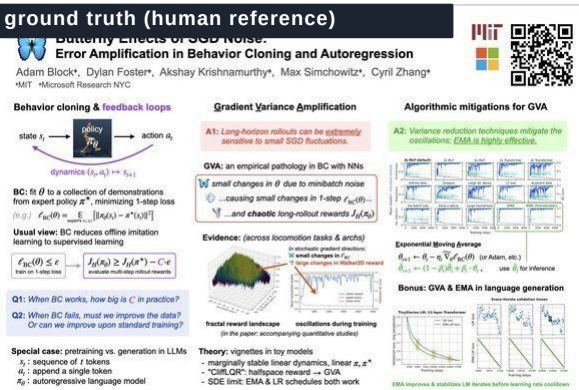
Aesthetic appeal is subjective and audience-dependent; the sub-criteria are scored by VLM judges that stand in for, but do not equal, human preference — a higher score means the raters prefer the poster, not that humans would. Comparison (i) also changes prompting, call count, and budget, so it does not isolate the loop; Codex at 3.27 (per judge 3.29 / 3.25) is second-highest in Table 1.

Source: arXiv:2607.04438 v2 (19 Jul 2026)

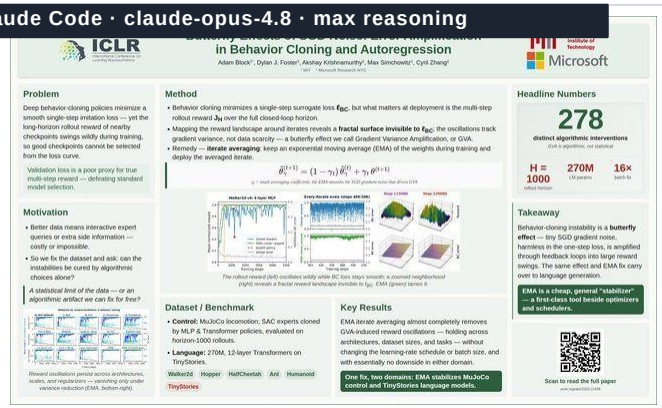
# With the skill, prompt, and pipeline fixed, five configurations each produce a full page differing in figure choice, density, and accent

Figure 10: the human reference and five runs of one skill under different harness, model, or reasoning effort

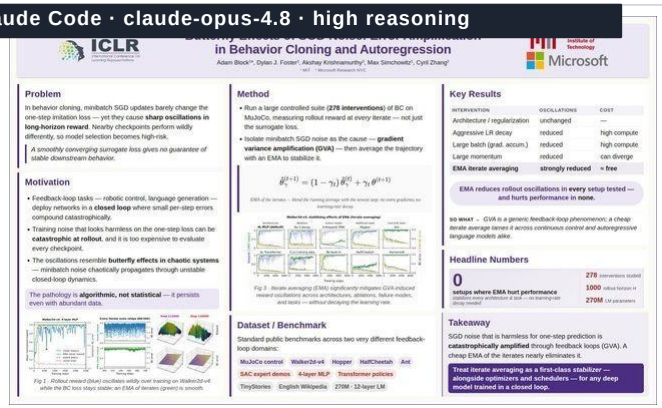
**Author ground truth (human reference)**



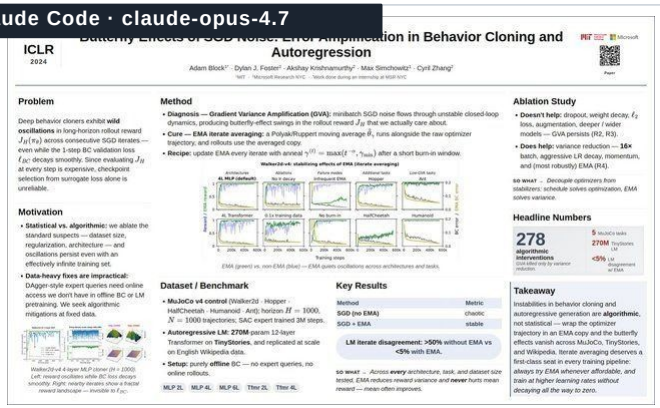

**Claude Code · claude-opus-4.8 · max reasoning**



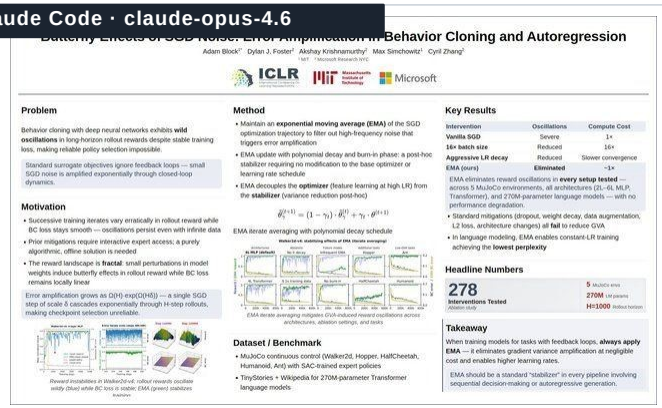

**Claude Code · claude-opus-4.8 · high reasoning**



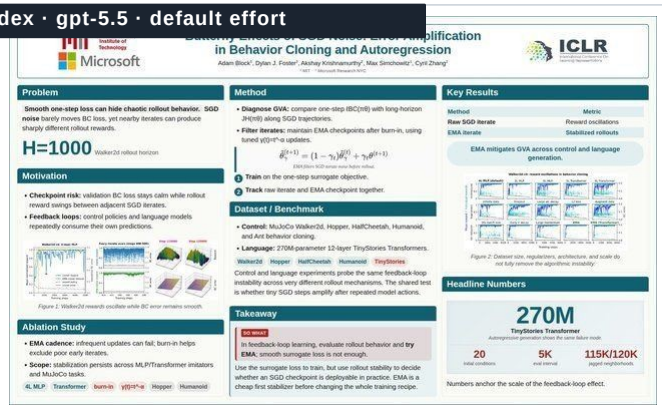

**Claude Code · claude-opus-4.7**




**Claude Code · claude-opus-4.6**

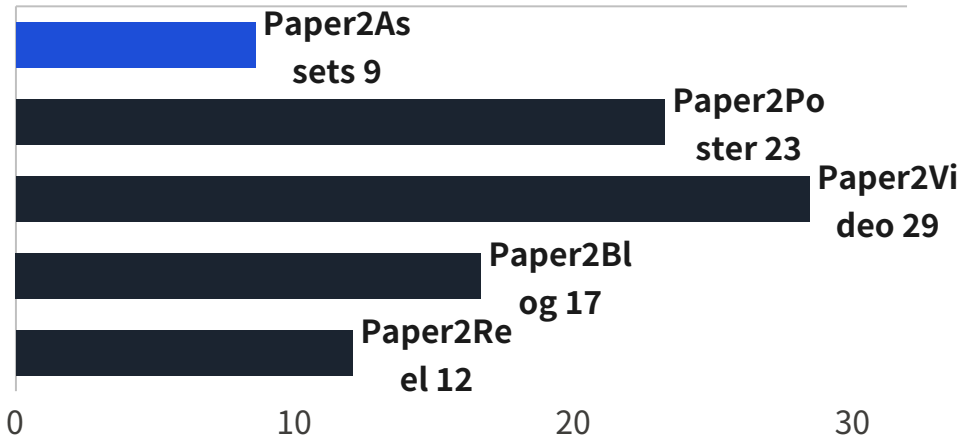



**Codex · gpt-5.5 · default effort**



# A full four-artifact bundle takes about 89 minutes of stage time and 2.6M input / 276K output tokens per paper

Table 4: per-stage time and tokens, mean over 5 papers on claude-opus-4-8; each skill runs as an isolated subprocess




| Skill | Heaviest stage | Minutes | Token share |
|---|---|---|---|
| Paper2Assets | **extract & figures** | 8.6 | 18% of input |
| Paper2Poster | **fill loop** | 14.8 | 18% of output |
| Paper2Video | **render & mux** | 9.2 | 6% of output |
| Paper2Blog | **QA gate & revision** | 10.0 | 17% of output |
| Paper2Reel | **QA gate** | 6.2 | 8% of output |

Input = distinct content tokens (fresh plus cache writes); Cache = cached-context re-reads, billed at about 10% of fresh input; Output = prorated generated tokens. Bars in the paper give each stage's share of the pipeline total.

## FULL PIPELINE PER PAPER

**89.2**
min summed stage time

**675**
turns

**2,568K**
input tokens

**108,546K**
cached re-reads (~10% of fresh)

**276K**
output tokens

- Paper2Poster and Paper2Video carry the largest generator subtotals: the measured-fill loop is Paper2Poster's longest stage; Paper2Video spends most of its time across deck construction, rendering and muxing, and QA. Paper2Blog is driven by its bilingual revision gate; Paper2Reel has the smallest subtotal.
- Running the three generators in parallel reduces elapsed wall-clock time; each skill's one-time context load (reading the paper and its figures) is folded into its first stage. When the LaTeX source is available the pipeline can read original figure files and captions instead of the PDF path.

# Three contexts span the dissemination spectrum on one pipeline: a camera-ready week, a lab's intake, a course's weekly packs

Who runs it, which artifacts they lean on, what they revise; Tables 2–3 audit what the video and blog generators emit

### 1 §4.1 The author's last mile

ONE-OFF, SINGLE PAPER

One Claude Code session per paper: one extraction pass populates the bundle, then the three generators emit an editable .pptx, a synchronized video.pptx + .mp4 pair, and a bilingual Word .docx pair the author hand-revises rather than regenerates — all navigable through one reel.html that doubles as a review surface for co-authors.

LEANS ON

- editable .pptx poster
- video.pptx + .mp4 pair
- bilingual .docx pair
- reel.html as a review surface

### 2 §4.2 Lab and org communication

CONTINUOUS, PER ACCEPTED PAPER

Wired into the publication-intake pipeline so every accepted paper auto-generates draft dissemination artifacts an editorial or communications team then polishes. The editable PowerPoint and Word outputs make it a draft-and-revise workflow; the bilingual blog covers a Chinese-WeChat and an English research-blog register from one evidence map.

LEANS ON

- draft artifacts per accepted paper
- PowerPoint and Word for hand-off
- one reel per paper for the comms team

### 3 §4.3 Educational reuse

PERIODIC, MANY-PAPER BATCH

A course, reading group, or public-understanding outlet feeds a reading list through the pipeline to produce paper-of-the-week briefing packs: a lecture deck, an asynchronous talk video, a newsletter blog. Instructors adjust framing or drop a slide; re-running on a corrected PDF re-issues the whole pack.

LEANS ON

- lecture deck
- asynchronous talk video
- newsletter blog
- versioned bundle per paper

## CAPABILITY AUDITS, TABLES 2 AND 3

- Table 2 (video) asks whether a system emits visual attention cues or cursor grounding, an editable PPTX deck, duration control, and narration audio; Table 3 (blog) asks about layout checks, embedded figures, editable DOCX output, and paper-input summarization.
- The audited products and research systems generally cover a single artifact; ResearchStudio-Reel emits all three, exposes the shared bundle, includes a native-editable source in each package, and stitches them into one surface. The audit documents breadth of the contract; it does not measure editing effort or the viewer's usability.

## GAP TO HUMAN POSTERS

Matching or exceeding the aggregate judge scores does not mean the output is on par with an author's own poster. A human designer routinely adds material the paper does not contain — a purpose-drawn method or overview diagram, explanatory icons — whereas the pipeline is grounded in the extracted bundle and reuses only figures and content that appear in the paper. Author taste also spans a wide, legitimate range; the pipeline analyses each paper's figure and content profile and chooses the layout that fits it, aiming for a sensible middle ground rather than any single house style.

Source: arXiv:2607.04438 v2 (19 Jul 2026)

# Five failure modes recur in end-to-end runs; the system is calibrated on ML / CV / NLP venues only; quantitative evaluation covers the poster

Each failure mode has a named mitigation; video and blog are compared on capability only; no human-rater study exists yet

| ID | Requirement | Solution the paper gives |
|---|---|---|
| L1 | | |
| L2 | | |
| L3 | | |
| L4 | | |
| L5 | | |

**Appendix B, provenance and disclosure**

A narrated video with burned-in subtitles is not obviously AI-produced at a glance. The authors recommend that deployments preserve provenance metadata — which skill and version, which bundle checksum, which theme, header, or TTS voice, which model — so readers can audit the generation chain and, where required, disclose AI involvement; the inventory manifest and the video QA record already carry this ancestry.

## DOMAIN SCOPE

Calibrated on ML, CV, and NLP venues, where paper layout, figure conventions, and the claim–evidence structure are relatively uniform. Transfer to biomedicine, physics, or design-heavy fields is untested and each skill's move catalogue may need extending; the primitives are domain-agnostic, but each venue family should be validated end-to-end before deployment.

## EVALUATION COVERAGE

Quantitative results are reported only for the poster, under two VLM judges on the Paper2Poster benchmark; the video benchmark of Paper2Video [4] is not applied. The audits in Tables 2–3 document output contracts but do not measure human editing effort, fidelity after edits, or whether section-level navigation improves understanding. RRP (§2.2) is an in-loop signal, not a headline metric; a third-party human-rater study across all three artifacts is left for future work.

## LICENSING, APPENDIX B

Per-figure copyright on extracted figures follows the source paper's license; logos follow Wikimedia Commons and Wikidata; the edge-tts client accesses Microsoft Edge's online service, so deployments verify the service terms. The pipeline redistributes neither figures nor logos: the user runs it on their own paper.

# Supported: a native-editable, section-aligned workspace with testable gates. Open: reading-and-recall evaluation and bespoke diagrams

From one-way automation to a workflow authors can inspect, navigate, and revise; the paper names the limits of its own proxies

## ESTABLISHED BY THE PAPER

1. **Five composable skills, one bundle, three native-editable artifacts**

   One Paper2Assets bundle feeds a native-editable PowerPoint poster and video deck and a bilingual Word blog; Paper2Reel binds their sections, slides, video times, captions, and passages.

2. **Posters judged above the authors' own on aesthetics**

   Best among automated systems on all three aesthetic sub-criteria and best or tied-best on two of three information sub-criteria under two VLM judges; 3.56 vs 3.03; overall wins on 74 and 95 of 100 papers.

3. **A pattern that may travel**

   The delivery-contract pattern may extend to other targets with native authoring formats, deterministic renderers, and testable package criteria — with its evaluative, generative, and architectural limits still to be studied (§6).

## TWO GAPS THAT MATTER MORE THAN COVERAGE

### Evaluation is proxy-bound

The aesthetic rubric and PaperQuiz pull in opposite directions — a denser poster wins comprehension while a cleaner one wins aesthetics — and neither measures whether a reader absorbs the work. The measured-fill loop optimizes a geometric density target, not understanding; the honest next step is to close the loop on a controlled human reading-and-recall signal rather than on a proxy a denser poster can always game.

### The generative gap

Because the pipeline reuses only figures that already exist in the paper, it cannot yet produce the bespoke method or overview diagrams human designers add to carry the narrative. Addressing this may require faithful figure synthesis, which reintroduces the hallucination risk the categorical gates were built to suppress, and would push gate discipline from layout onto the factual content of generated visuals.

THE PAPER'S CLOSING SENTENCE, §7

“By keeping generated artifacts editable and connected after generation, ResearchStudio-Reel shifts the last mile from one-way automation toward a workflow authors can inspect, navigate, and revise.”

# Sources

Everything shown is quoted from arXiv:2607.04438 v2; the figures are the paper's own

**PAPER**

ResearchStudio-Reel: Automate the Last Mile of Research from Paper to Poster, Video, and Blog — arXiv:2607.04438 v2, 19 July 2026

**PROJECT**

ResearchStudio project page

**CODE**

microsoft/ResearchStudio on GitHub — MIT license; each skill's SKILL.md is the source of truth for its workflow

## PROVENANCE OF THIS DECK

- Figures 1–10 are reproduced from the paper at their extracted resolution; none is redrawn.
- Table 1 and Table 4 values are quoted as printed; per-judge scores are in Appendix D (Table 5); the three single-shot baselines share one fixed prompt reproduced in Appendix G.
- Titles are this deck's readings of the paper's claims; observation, inference, and speculation keep the paper's wording.

## LICENSING NOTES STATED BY THE PAPER

- Extracted paper figures follow the source paper's license; the pipeline itself does not redistribute paper figures or logos.
- Institution and venue logos fetched by Paper2Assets follow Wikimedia Commons and Wikidata licenses.
- The edge-tts client accesses Microsoft Edge's online text-to-speech service; deployments should verify the applicable service terms.